

Advanced Bayesian Learning

Variational inference

Lecture 6

Mattias Villani

Department of Statistics
Stockholm University



Fixed form VI

- The independence assumption in **mean field VI is restrictive**.
- Blocking parameters as in structural VI often helps a lot.
- **Fixed form VI:**
 - ▶ Assume parametric form $q_{\lambda}(\boldsymbol{\theta})$ with hyperparameters λ
 - ▶ Minimize $\text{KL}[q_{\lambda}(\boldsymbol{\theta}) \parallel p(\boldsymbol{\theta}|\mathbf{y})]$ wrt λ .
- As before, we actually maximize $\text{ELBO}(\lambda)$.
- Ex 1: $q_{\lambda} = N(\boldsymbol{\theta}|\boldsymbol{\mu}, \Sigma)$, $\lambda = (\boldsymbol{\mu}, \mathbf{L})$, with Cholesky $\Sigma = \mathbf{L}\mathbf{L}^T$.
- Ex 2: $q_{\lambda} = N(\boldsymbol{\theta}|\boldsymbol{\mu}, \mathbf{a}\mathbf{a}^T + \mathbf{D})$, $\lambda = (\boldsymbol{\mu}, \mathbf{a}, \mathbf{d})$, where $\mathbf{a}$ is a vector, $\mathbf{D} = \text{Diag}(\mathbf{d})$.
- Ex 3: q_{λ} is a copula, mixture of normals etc etc
- From now on: vectors will not be **bold**.

Fixed form VI - Gradient based optimization

- **Gradient ascent:** optimize $\text{ELBO}(\lambda)$ wrt λ using step size $a > 0$

for $t = 1, 2, \dots$ until convergence do:

$$\lambda^{(t+1)} = \lambda^{(t)} + a \cdot \nabla_{\lambda} \text{ELBO}(\lambda^{(t)})$$

- Stop when changes in $\text{ELBO}(\lambda^{(t)}) < \epsilon$.

- **Lower Bound**

$$\text{ELBO}(\lambda) = \mathbb{E}_{q_{\lambda}} \left[\log \frac{p(y|\theta)p(\theta)}{q_{\lambda}(\theta)} \right] = \int q_{\lambda}(\theta) h_{\lambda}(\theta) d\theta$$

$$h_{\lambda}(\theta) := \log \frac{p(y|\theta)p(\theta)}{q_{\lambda}(\theta)} = \log p(y|\theta)p(\theta) - \log q_{\lambda}(\theta).$$

Gradient of LB is an expectation wrt $q_\lambda(\theta)$

$$\begin{aligned}\nabla_\lambda \text{ELBO}(\lambda) &= \int \nabla_\lambda (q_\lambda(\theta) h_\lambda(\theta)) d\theta \text{ [interchange } \int \text{ and } \nabla_\lambda] \\ &= \int ((\nabla_\lambda q_\lambda(\theta)) h_\lambda(\theta) + q_\lambda(\theta) (\nabla_\lambda h_\lambda(\theta))) d\theta \text{ [product rule]}\end{aligned}$$

Since $\nabla_\lambda \log q_\lambda(\theta) = \nabla_\lambda q_\lambda(\theta) / q_\lambda(\theta)$ we have $\nabla_\lambda q_\lambda(\theta) = q_\lambda(\theta) \nabla_\lambda \log q_\lambda(\theta)$

$$\nabla_\lambda h_\lambda(\theta) = \nabla_\lambda (\log p(y|\theta)p(\theta) - \log q_\lambda(\theta)) = -\nabla_\lambda \log q_\lambda(\theta) = -\frac{\nabla_\lambda q_\lambda(\theta)}{q_\lambda(\theta)}$$

Using that $\int q_\lambda(\theta) \left(\frac{\nabla_\lambda q_\lambda(\theta)}{q_\lambda(\theta)} \right) d\theta = \int \nabla_\lambda q_\lambda(\theta) d\theta = \nabla_\lambda \int q_\lambda(\theta) d\theta = \nabla_\lambda 1 = 0$

$$\begin{aligned}\nabla_\lambda \text{ELBO}(\lambda) &= \int ((\nabla_\lambda q_\lambda(\theta)) h_\lambda(\theta) + q_\lambda(\theta) (\nabla_\lambda h_\lambda(\theta))) d\theta \\ &= \int q_\lambda(\theta) (\nabla_\lambda \log q_\lambda(\theta)) h_\lambda(\theta) d\theta - \int q_\lambda(\theta) \left(\frac{\nabla_\lambda q_\lambda(\theta)}{q_\lambda(\theta)} \right) d\theta \\ &= \int q_\lambda(\theta) (\nabla_\lambda \log q_\lambda(\theta)) h_\lambda(\theta) d\theta \\ &= \mathbb{E}_{q_\lambda} (h_\lambda(\theta) \nabla_\lambda \log q_\lambda(\theta))\end{aligned}$$

VI - Stochastic gradient ascent

- Gradient of LB is an expectation wrt $q_\lambda(\theta)$

$$\nabla_\lambda \text{ELBO}(\lambda) = \mathbb{E}_{q_\lambda} [\nabla_\lambda \log q_\lambda(\theta) \times h_\lambda(\theta)]$$

- Monte Carlo: simulate $\theta^{(1)}, \dots, \theta^{(S)} \sim q_\lambda(\theta)$ and estimate

$$\nabla_\lambda \widehat{\text{ELBO}}(\lambda) = \frac{1}{S} \sum_{s=1}^S \nabla_\lambda \log q_\lambda(\theta_s) \times h_\lambda(\theta_s)$$

Algorithm 1: Basic FFVB algorithm

Input: Initial value $\lambda^{(0)}$, tolerance ϵ , step sequence $\{a_t\}_{t=0}^\infty$

for $t = 0, 1, \dots$ *until convergence* **do**

- Generate $\theta_s \sim q_{\lambda^{(t)}}(\theta)$, for $s = 1, \dots, S$

- Estimate the LB gradient unbiasedly

$$\nabla_\lambda \widehat{\text{LB}}(\lambda) = \frac{1}{S} \sum_{s=1}^S \nabla_\lambda \log q_\lambda(\theta_s) \times h_\lambda(\theta_s)|_{\lambda=\lambda^{(t)}}$$

- Update

$$\lambda^{(t+1)} = \lambda^{(t)} + a_t \cdot \nabla_\lambda \widehat{\text{LB}}(\lambda^{(t)})$$

end

Output: Optimal λ

- In the sequel, ELBO is sometimes written as LB.

Monitoring convergence

- Sufficient conditions for convergence:

$$a_t > 0, \quad \sum_t a_t = \infty \quad \text{and} \quad \sum_t a_t^2 < \infty$$

- Example:

$$a_t = \begin{cases} \epsilon_0 & \text{if } t \leq \tau \\ \epsilon_0 \frac{\tau}{t} & \text{if } t > \tau \end{cases}$$

- Hard to monitor convergence on $\widehat{\text{LB}}(\lambda^{(t)})$ since it is noisy.

- Check convergence on local average:

$$\overline{\text{LB}}(\lambda^{(t+1)}) = t_W^{-1} \sum_{k=1}^{t_W} \text{LB}(\widehat{\lambda^{(t-k+1)}})$$

- MNT: $t_W = 20$ or $t_W = 50$ and tolerance $\epsilon = 10^{-5}$ common.

Adaptive learning rate

- **Learning rate** a_t should be small when $\mathbb{V} \left(\nabla_{\lambda} \widehat{\text{LB}}(\lambda^{(t)}) \right)$ is large, otherwise optimizer may backtrack.
- Algorithm above used **same learning rate** a_t for all λ_k .
- But $\mathbb{V} \left(\nabla_{\lambda_k} \widehat{\text{LB}}(\lambda^{(t)}) \right)$ may vary with different λ_k .
- Scale gradients with moving average of $\mathbb{V} \left(\nabla_{\lambda_k} \widehat{\text{LB}}(\lambda^{(t)}) \right)$
- Let $g_t = \nabla_{\lambda} \widehat{\text{LB}}(\lambda^{(t)})$ and $v_t = g_t^2$ (elementwise, i.e. $g_t \odot g_t$).

Stochastic gradient ascent with adaptive gradients

Algorithm 2: FFVB algorithm with adaptive gradients

Input: Initial value $\lambda^{(0)}$, tolerance ϵ , step sequence $\{\alpha_t\}_{t=0}^{\infty}$, unbiased gradient estimator g , g_0 and v_0 , β_1 and β_2 .

Set $\bar{g} \leftarrow g_0$ and $\bar{v} \leftarrow v_0$

for $t = 0, 1, \dots$ *until convergence* **do**

- $\bar{g} \leftarrow \beta_1 \bar{g} + (1 - \beta_1) g_t$
- $\bar{v} \leftarrow \beta_2 \bar{v} + (1 - \beta_2) g_t^2$ [elementwise]
- $\lambda^{(t+1)} = \lambda^{(t)} + \alpha_t \cdot \bar{g} / \sqrt{\bar{v}}$ [elementwise]

end

Output: Optimal λ

Natural gradient

- Same distance in λ -space can give very different changes in $\text{KL}(q_\lambda \| p(\theta|y))$ depending on the geometry of $q_\lambda(\theta)$.
- Example: changing the mean of $q_\lambda(\theta)$ can have very different effect on $\text{KL}(q_\lambda \| p(\theta|y))$ depending on the variance of $q_\lambda(\theta)$.
- The **natural gradient** solves this:

$$\nabla_\lambda \text{LB}(\lambda)^{\text{nat}} = I_F^{-1}(\lambda) \nabla_\lambda \text{LB}(\lambda)$$

where $I_F(\lambda) = \mathbb{V}_{q_\lambda}(\nabla_\lambda \log q_\lambda(\theta))$ is the Fisher Information.

- Compute inverse by iterative conjugate gradient:
Solve approximately $I_F(\lambda)x = \nabla_\lambda \text{LB}(\lambda)$ for x .
- In exponential families, $\nabla_\lambda \text{LB}(\lambda)^{\text{nat}}$ is simpler than $\nabla_\lambda \text{LB}(\lambda)$, see Blei et al. (2017).

Variance reduction by control variates

- The **variance of gradient estimator**

$$\widehat{\nabla_{\lambda} \text{LB}(\lambda)} = \frac{1}{S} \sum_{s=1}^S \nabla_{\lambda} \log q_{\lambda}(\theta_s) \times h_{\lambda}(\theta_s)$$

is often large. Problematic for stochastic gradient ascent.

- Estimator with **control variates** c_k

$$\widehat{\nabla_{\lambda_k} \text{LB}(\lambda)} = \frac{1}{S} \sum_{s=1}^S \nabla_{\lambda_k} \log q_{\lambda}(\theta_s) \times (h_{\lambda}(\theta_s) - c_k)$$

- **Unbiased** since $\mathbb{E}(\nabla_{\lambda} \log q_{\lambda}(\theta)) = 0$. **Lower variance**.
- **Optimal** c_k that minimizes $\mathbb{V}(\widehat{\nabla_{\lambda_k} \text{LB}(\lambda)})$ derived in MNT.
- **Optimal** c_k estimated in gradient ascent. Algorithm 3, MNT.

The reparametrization trick

- Suppose we can generate $\theta \sim q_\lambda(\cdot)$ by generating $\varepsilon \sim p_\varepsilon(\cdot)$ and the deterministic transformation $\theta = g(\lambda, \varepsilon) \sim q_\lambda(\cdot)$.
- Ex: $q_\lambda(\cdot) = N(\mu, \Sigma)$, then $g(\lambda, \varepsilon) = \mu + \Sigma^{1/2}\varepsilon$, $\varepsilon \sim N(0, I)$.
- $\text{LB}(\lambda)$ can be expressed as an expectation wrt $\varepsilon \sim p_\varepsilon(\cdot)$

$$\text{LB}(\lambda) = \mathbb{E}_{q_\lambda} (h_\lambda(\theta)) = \mathbb{E}_{p_\varepsilon} (h_\lambda(g(\lambda, \varepsilon)))$$

- The **gradient** is an expectation wrt $\varepsilon \sim p_\varepsilon(\cdot)$

$$\nabla \text{LB}(\lambda) = \mathbb{E}_{p_\varepsilon} (\nabla_\lambda g(\lambda, \varepsilon) \nabla_\theta h_\lambda(\theta))$$

since $\mathbb{E}_{p_\varepsilon} (\nabla_\lambda h_\lambda(\theta)) = 0$ (MNT).

- **Unbiased estimator** by generating $\varepsilon_1, \dots, \varepsilon_S \stackrel{iid}{\sim} p_\varepsilon(\cdot)$

$$\widehat{\nabla_\lambda \text{LB}(\lambda)} = \frac{1}{S} \sum_{s=1}^S \nabla_\lambda g(\lambda, \varepsilon_s) \nabla_\theta h_\lambda(g(\lambda, \varepsilon_s))$$

- Lower variance since uses gradient information $\nabla_\theta h_\lambda(\theta)$.¹

¹Xu et al (2019). Variance Reduction Properties of the Reparameterization Trick. AISTats.